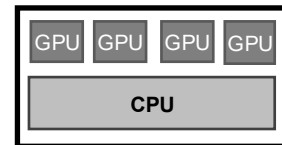


NEKRS-ML DATA TRANSFER: OVERVIEW

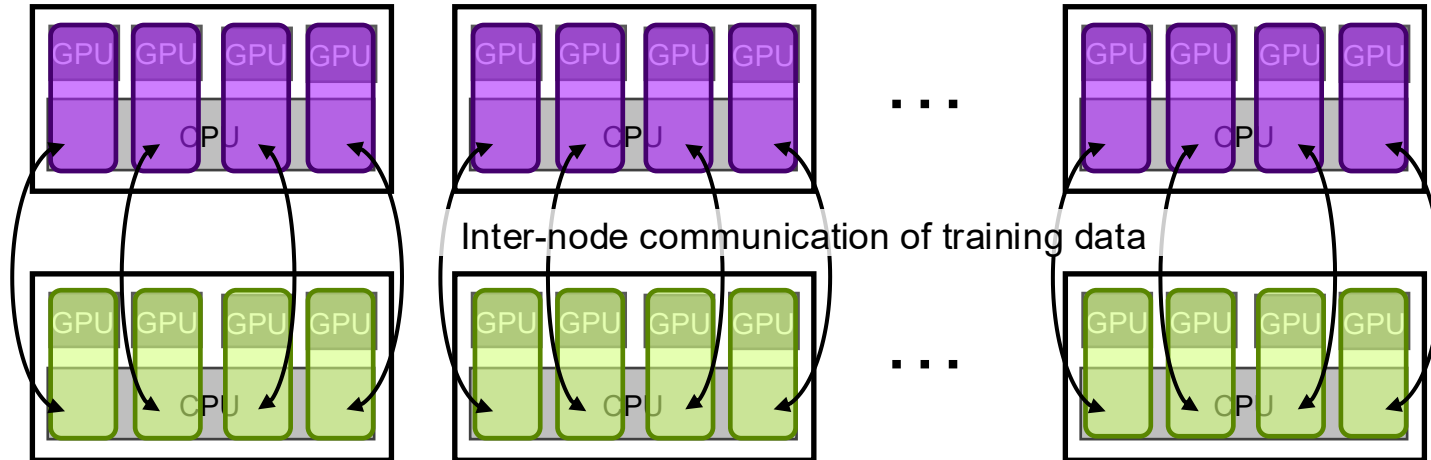
- The online data transfer component of the workflow is implemented using ADIOS2 SST engine (RDMA or MPI transport layer preferred)
- Benchmark to be run in clustered mode to force inter-node communication
 - nekRS and Dist-GNN training run on distinct set of nodes
 - N-N transport pattern across system high-speed interconnect

Example of heterogeneous node



nekRS 
Dist-GNN 
Data transfer 

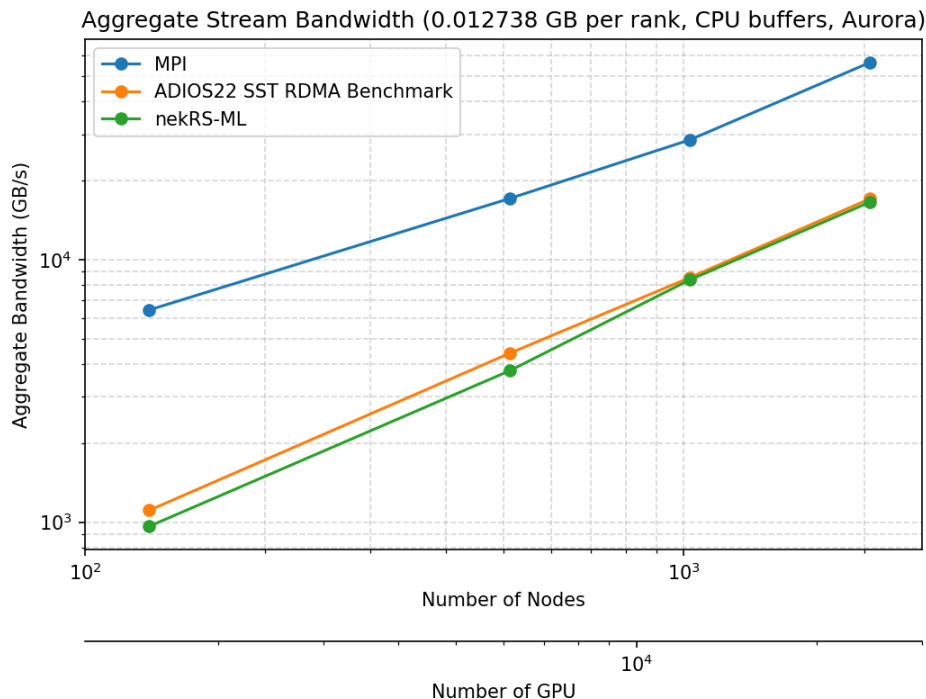
Clustered Deployment



WORKFLOW DATA TRANSFER: PERFORMANCE

- Clustered deployment at full system scale limited by bisection bandwidth of high-speed interconnect
 - For homogeneous system, tests full-scale bisection bandwidth
 - For macro-heterogeneous system (separate ModSim and AI partitions), tests bisection bandwidth between partitions²
- nekRS-ML workflow uses CPU buffers for data transfer
 - OCCA device arrays containing velocity field are moved to host before being passed to ADIOS2 SST stream write
 - Host offload is NOT included in data transfer FOM measurement

WORKFLOW DATA TRANSFER: PERFORMANCE



See [this](#) repo for descriptions of the MPI and ADIOS2 SST RDMA benchmark used for comparison.